Bitwise XOR Operation ($a \oplus b$)

Programming Fundamentals

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Problem Overview: Calculating a XOR b

To compute the bitwise Exclusive OR (XOR) of two numbers, we can represent the logic in three standard ways:

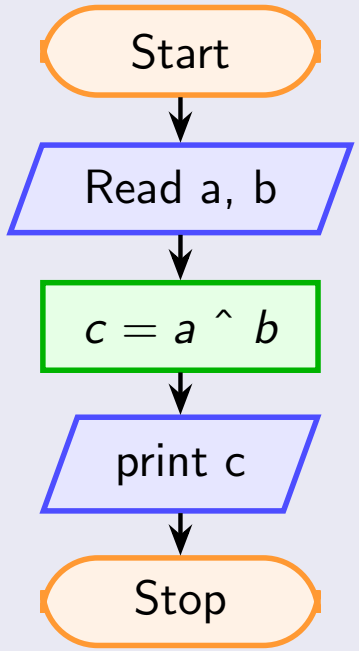
- **Algorithm:** Step-by-step logic.
- **Flowchart:** Graphical representation using input, processing, and output blocks.
- **Program:** Concrete implementation using the C language ^ operator.

1. Algorithm & 2. Flowchart

Algorithm Steps

1. **Start**
2. **Read** integers a and b
3. **Calculate** $c = a \oplus b$
4. **Print/Display** c
5. **Stop**

Visual Flowchart



3. Program Implementation (C Language)

Source Code using the ^ Operator

```c
#include <stdio.h>

int main() {
    int a, b, c;

    printf("Enter value of a: ");
    scanf("%d", &a);

    printf("Enter value of b: ");
    scanf("%d", &b);

    // ^ is the bitwise XOR operator in C
    c = a ^ b;

    printf("Result of a XOR b is: %d\n", c);

    return 0;
}
```